Rezk Youssef Rezk

Data Analyst — Task Engineer

Cairo, Egypt | +20101 748 6134 | rezq.yussef@gmail.com
LinkedIn | GitHub | Portfolio

SUMMARY

Motivated Computer Science student and Data Analyst with a strong foundation in algorithms and complex problem-solving (ICPC). Proficient in Python (Pandas, NumPy) for advanced data manipulation, statistical computation, and handling messy datasets. Experienced in simplifying complex technical logic and writing precise verification code as a C++ Instructor. Comfortable working independently in remote Linux/Ubuntu environments. Eager to design analytical tasks and apply machine learning knowledge to drive data-driven insights. Currently enhancing expertise through the Digital Egypt Pioneers Initiative (DEPI) training.

EDUCATION

High Future Institute for Specialized Technological Studies (Future Academy) — Cairo, Egypt
B.Sc. in Computer Science (GPA: 3.44) — Jan 2023 – Present

- Focused on core Computer Science fundamentals including Data Structures, Algorithms, and Object-Oriented Programming (OOP).
- Gained a strong mathematical foundation through coursework in Linear Algebra, Probability, Statistics, and Numerical Analysis, essential for Data Analysis and evaluating analytical conclusions.
- Actively applying theoretical knowledge through competitive programming (ICPC) and practical software projects.

EXPERIENCE

Data Science Track — Digital Egypt Pioneers Initiative (DEPI) — Cairo, Egypt
Trainee / Participant — Nov 2025 – Jul 2026

- Mastered data analysis and visualization workflows using Python, Pandas, NumPy, and Matplotlib.
- Developed automated data collection scripts using Web Scraping and APIs, integrated with SQL for database management.
- Selected for an intensive scholarship by the Ministry of Communications (MCIT) to apply advanced data science techniques and statistical concepts to real-world datasets.

C++ Instructor — ICPC Community @ Future Academy — Cairo, Egypt
Technical Instructor — Oct 2025 – Present

- Instructed students in Advanced C++ and Data Structures & Algorithms to prepare for competitive programming contests.
- Authored, designed, and tested custom algorithmic problems to simulate official contest environments, requiring precise logic and verification scripts to assess students' solutions.

PROJECTS

Retail Sales Data Cleaning & Verification — **Jun 2026**
Complex Data Analysis & Validation — *Python, Pandas, NumPy*

- Engineered a data cleaning pipeline to process messy retail datasets, resolving structural anomalies, duplicates, and missing values.
- Implemented advanced mathematical imputation to accurately deduce missing numerical data (Price, Quantity, Total) and utilized smart vectorized mapping for categorical deduction.

- Wrote precise business logic validation scripts using NumPy to programmatically cross-reference and verify mathematical consistency across thousands of transaction records.
- Optimized memory consumption by converting high-cardinality text to categorical types and engineered temporal features (Year, Month, Day) from datetime objects.

Diabetes Prediction (AI) — April 2026

Deep Learning — *Python, TensorFlow, Keras, Scikit-Learn*

- Developed a Feedforward Neural Network (FNN) model to predict the early onset of diabetes based on medical records.
- Built and optimized the deep learning architecture using TensorFlow and Keras, achieving a high testing accuracy of 94% with robust precision and recall.
- Handled data preprocessing, missing values imputation, and feature scaling using Pandas and Scikit-Learn.

Sales Performance Dashboard — May 2026

Data Analysis & Visualization — *Python, Plotly Dash, Pandas, SQL*

- Developed an interactive business intelligence tool using Python and Plotly Dash for real-time monitoring of retail performance.
- Aggregated, cleaned, and processed complex data from CSV and SQL sources using Pandas to calculate specific Key Performance Indicators (KPIs) including Total Revenue, Sales volume, and Average Order Value.
- Applied statistical concepts to analyze distributions and track performance metrics across multiple product categories (Bed, Chair, Desk, Sofa, Table) to provide verifiable analytical conclusions.

Smart Home Automation System — Apr 2026

IoT & Embedded Systems — *C++, ESP32, FreeRTOS*

- Developed an IoT-based home automation system using ESP32, Arduino microcontrollers, and FreeRTOS.
- Programmed efficient C++ firmware for real-time sensor data processing and hardware control.
- Implemented wireless connectivity to enable seamless automation and remote device management via a custom web interface.

Employee Attrition Prediction & Analysis — Feb 2026

Machine Learning & Statistics — *Python, Scikit-learn, EDA*

- Implemented a machine learning project using Python to analyze and predict employee attrition rates.
- Handled real-world dataset cleaning, exploratory data analysis (EDA), and identified critical correlations and outliers within the workforce data.

SKILLS

Data Analysis & AI: Pandas, NumPy, BeautifulSoup (Web Scraping), Statistical Computation, Data Validation & Verification, TensorFlow, Keras.

Programming Languages: C++, Python, SQL, JavaScript.

Tools & Environments: Git, GitHub, VS Code, Microsoft Excel, Linux (Ubuntu), Docker (Familiarity).

Soft Skills: Problem Solving, Technical Training, Communication, Remote Independent Work.

Languages: English, Arabic.